

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2019-P2-Model)** **Date: 22-03-2026**

Time: 02.30Pm to 05.30Pm **GTA-19** **Max. Marks: 186**

22-03-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2019-P2)_GTA-19_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

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Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2019-P2-Model

Time: 3:00Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 186**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – II(Q.N : 9 – 14)	Questions with Numerical Value Answer Type	3	0	6	18
Sec – III(Q.N : 15 – 18)	Matching Type	3	-1	4	12
Total				**18**	**62**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – II(Q.N : 27 – 32)	Questions with Numerical Value Answer Type	3	0	6	18
Sec – III(Q.N : 33 – 36)	Matching Type	3	-1	4	12
Total				**18**	**62**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – II(Q.N : 45 – 50)	Questions with Numerical Value Answer Type	3	0	6	18
Sec – III(Q.N : 51 – 54)	Matching Type	3	-1	4	12
Total				**18**	**62**

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PHYSICS **Max Marks: 62**

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 1 (Max Marks: 32)

- Section 1 contains eight questions.
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate the Answer to each question:
  Full Marks: +4 (If the answer is correct)
  Partial Marks: +3 (If all the four options are correct but only 3 are chosen)
  Partial Marks: +2 (If THREE or More options are correct but only 2 correct answers are chosen)
  Partial Marks: +1 (If TWO or More options are correct but only 1 correct answer is chosen)
  Zero Marks: 0 (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

1. Two satellites, each of mass m, revolve around a planet of mass $M \gg m$ in circular orbits of radii R_1 and R_2 where $R_2 = 2R_1$.

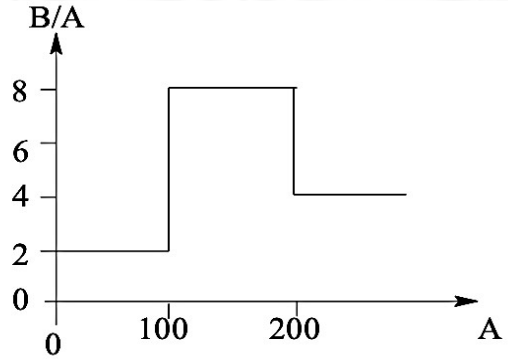
The satellites are connected by a light, inextensible, tensioned cable of length R_1. Because of this constraint, both satellites complete one revolution in the same orbital period T.

Let v_1, v_2 be the actual orbital speeds of the satellites in the presence of the cable. Let v_1', v_2' be the speeds they would have if no cable were present (i.e., if each orbited freely at radii R_1 and R_2).

The correct statements are:

A) $v_1 < v_1'$ **B)** $v_2 > v_2'$ **C)** $\frac{v_1}{v_1'} = \sqrt{\frac{5}{12}}$ **D)** $\frac{v_2}{v_2'} = \sqrt{\frac{10}{3}}$

2. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in figure below. Use this plot shown in figure to choose the correct statements from the following.



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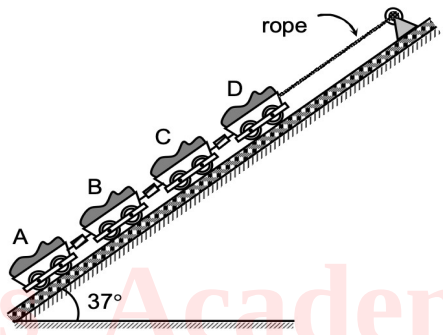
A) Fusion to two nuclei with mass numbers lying in the range of $1 < A < 50$ will release energy.

B) Fusion of two nuclei with mass numbers lying in the range of $51 < A < 100$ will release energy.

C) Fission of a nucleus lying in the mass range of $100 < A < 200$ will release energy when broken into two equal fragments.

D) Fission of a nucleus lying in the mass range of $200 < A < 260$ will release energy when broken into two equal fragments.

3. Four ore wagon cars are held at rest on a frictionless inclined of 37° with the help of an extensible rope as shown in the figure. Any two cars are connected with an inextensible coupling. The extension in the rope is 20 cm in equilibrium and mass of each ore wagon car is 1000 kg. Now if middle coupling (between wagon B and C) is broken due to some problem then choose the correct options. ($g = 10$ m/s^2)



A) The acceleration of cars 'A' and 'B' is 6 m/s^2

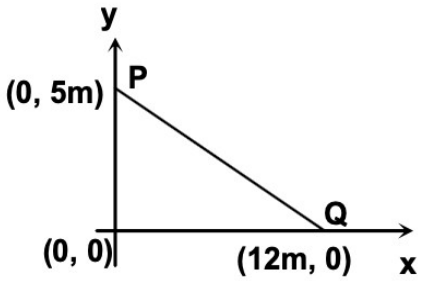
B) The frequency of cars 'C' and 'D' is $\frac{\sqrt{60}}{2\pi}$ Hz.

C) The amplitude of oscillations of cars C and D is 10 cm.

D) The amplitude of oscillations of cars C and D is 20 cm.

4. A particle is executing simple harmonic motion along a line $12y + 5x = 60$ only in between the extreme points 'P' and 'Q'. At $t = 0$, the particle starts its motion from the extreme point 'P'. The time period of simple harmonic motion executed by the particle is 8 sec. Then choose the correct option(s).

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A) The position vector of the particle at time t is given by

$$\vec{r}=\left[12\sin^2\left(\frac{\pi t}{8}\right)\hat{i}+5\cos^2\left(\frac{\pi t}{8}\right)\hat{j}\right]\text{m}.$$

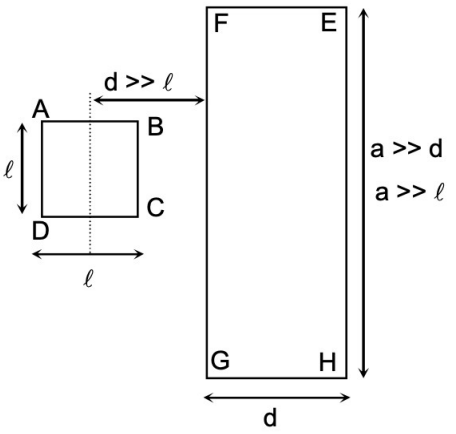
B) The position vector of the particle at time t is given by $\vec{r}=\left[12\cos^2\left(\frac{\pi t}{8}\right)\hat{i}+5\sin^2\left(\frac{\pi t}{8}\right)\hat{j}\right]\text{m}.$

C) The velocity vector of the particle at time t is given by

$$\vec{v}=\left[\frac{3\pi}{2}\sin\left(\frac{\pi t}{4}\right)\hat{i}-\frac{5\pi}{8}\sin\left(\frac{\pi t}{4}\right)\hat{j}\right]\text{m/s}.$$

D) The amplitude of simple harmonic motion executed by the particle is $\frac{13}{2}$ m.

5. A square conducting loop ABCD and a rectangular conducting loop EFGH are placed on the same plane as shown in the figure. The loop ABCD is placed symmetrically with respect to EFGH such that its central line is at a distance d from the loop EFGH. Side length 'ℓ' of small loop ABCD is much smaller than the length 'a' and width 'd' of the bigger loop EFGH. A source is applied on loop ABCD, such that it sets up a current $I=kt$ in it. Here k is a positive constant and t is time. Then choose the correct option(s).



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A) If the current in the loop ABCD is in clockwise sense then the induced current in the loop EFGH due to this current is in anticlockwise sense.

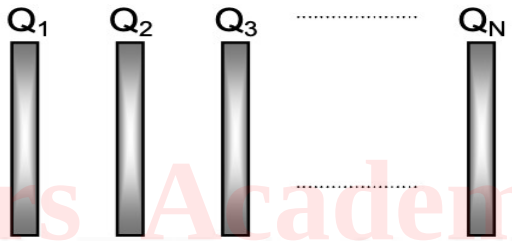
B) If the current in the loop ABCD is in clockwise sense then the induced current in the loop EFGH due to this current is in clockwise sense.

C) emf induced in the loop EFGH due to current in the loop ABCD is $\frac{\mu_0 \ell^2 k}{4\pi d}$.

D) emf induced in the loop EFGH due to current in the loop ABCD is $\frac{3\mu_0 \ell^2 k}{4\pi d}$.

6. N large square metallic plates are separated by a small distance and arranged parallel to its nearby plates as shown in the figure. Each of the plates are imparted charges $Q_1, Q_2 \ldots\ldots\ldots\ldots Q_N$ respectively.

Choose the correct answer(s) from the options.



A) Charges on the outermost surfaces of the plate system are equal and opposite.

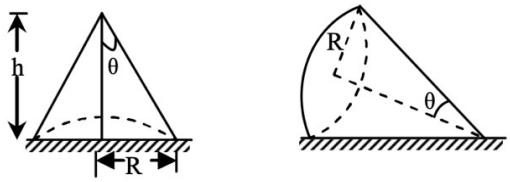
B) If the plate 1 is earthed, then charge flown to the earth is Q_1.

C) If the plate 3 is earthed, then charge flown to the earth is $\frac{\sum_{i=1}^{N} Q_i}{N}$

D) If the plate 4 is earthed, the charge flown to the earth is $\sum_{i=1}^{N} Q_i$.

7. A solid cone is placed on a horizontal surface with height 'h' and radius 'R'. Its apex angle is 'θ'. To change the position of the cone from figure (A) to figure (B), no work is required, then

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A) $\theta = \sin^{-1}\left(\frac{1}{3}\right)$ **B)** $\theta = \frac{\pi}{4}$ **C)** $\frac{h}{R} = 1$ **D)** $\frac{h}{R} = 2\sqrt{2}$

8. A vertical hand-pump contains air which is initially at atmospheric pressure p_0 and temperature T_0. The outlet valve of the pump is closed. A student Jatin pushes the piston in and compresses the trapped air quasistatically (slowly) with a constant force until the pressure inside becomes p and the air temperature becomes T.

He then connects a balloon to the outlet and opens the valve. The balloon starts to inflate slowly, while Jatin keeps pressing the piston with the same constant force as before.

Assume:

- Balloon is pumped completely full.
- heat exchange with surroundings and with the balloon wall is negligible
- stretching work of the balloon rubber is negligible,
- pressure inside the balloon is always equal to the outside air pressure p_0
- the air behaves as an ideal gas with molar heat capacity at constant volume $C_V = 3R/2$

Choose correct option(s)

A) The final temperature of the air in the balloon is equal to the air temperature in the pump just before opening the valve, that is $T_2 = T$

B) To press the air located outside the balloon, the gas inside the balloon does the work $= p_0 V_2$.

C) During balloon inflation, the total internal energy of the gas increases.

D) The final temperature of the air in the balloon is equal to $T_2 = 3\ T\ /\ 2$

SECTION – II
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**

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Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 2 (Max Marks: 18)

- Section two contains six (6) questions. The answer to each question is a NUMERICAL VALUE
- For each question enter the correct numerical value of the answer using the mouse and the on-screen virtual numerical keypad in the place designated to enter the answer. If the numerical values has more than two decimal places, truncate/round off the value to TWO decimal places.

Answer to each question will be evaluated according to the following marking scheme

Full Marks: +3 (If only the correct numerical value is entered)

Zero Marks: 0 (In all other cases)

9. A dielectric rod of mass m and length L carries equal positive charges q at its two ends. The rod is constrained to move only along the x-axis (no rotation).

A non-uniform electric field exists such that:

- For $x<0, \vec{E}=+E_0\hat{i}$
- For $x>0, \vec{E}=-E_0\hat{i}$

Initially, the center of the rod is at the origin and the rod is along the x-axis. The rod is given an initial velocity v along the positive x-direction.

The period of the oscillations is $T=\frac{xL}{v}+\frac{ymv}{E_0q}$. Fill the value of $x+y$.

10. A sealed water bottle of volume 2.0 L is left inside a car during cold weather.

The temperature inside the car reaches -3°C. The water inside the bottle, however, does not freeze because very pure and undisturbed water can remain in a super cooled liquid state even below 0°C.

When the bottle is shaken, nucleation occurs and a certain amount of water freezes instantaneously, while the rest remains liquid.

The properties of water are:

- Specific heat of liquid water: $c=4200\ \text{J kg}^{-1}{}^{\circ}\text{C}^{-1}$
- Density of water: $\rho=1000\ \text{kg m}^{-3}$
- Latent heat of fusion of ice: $\lambda=340\ \text{kJ kg}^{-1}$

Assume that no heat exchange occurs with the surroundings during the process.

Determine the mass of ice formed in the bottle.

Report your answer in grams.

11. A thin-walled glass capillary tube of inner radius r_1 is inserted coaxially inside another thin-walled glass tube of inner radius r_2. Both tubes are then placed vertically in water. Assuming complete wetting and negligible tube thickness and neglect end correction due to the meniscus. determine the value of r_1/r_2 for which the water rises to the **same height** in both tubes due to capillarity.

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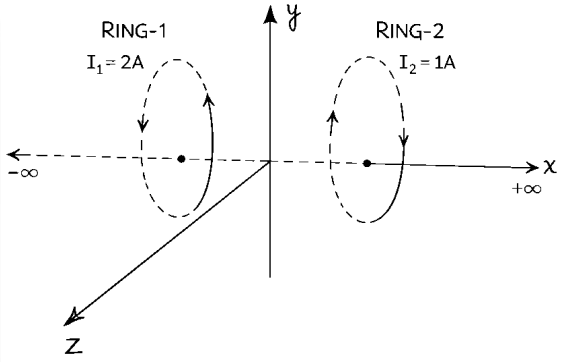
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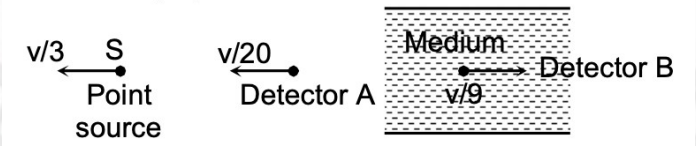
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12. Two co-axial rings of same radius $R = 10$ cm are placed parallel to the y-z plane, such that x-axis is the axis of the rings. Ring 1 carries a current of 2 Amp and ring 2 carries a current of 1 Amp as shown in the figure. The separation between the rings is $d = 50$ cm.



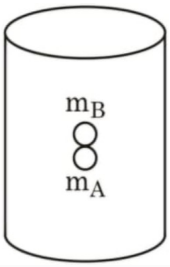
Find the magnitude of $\dfrac{\int_{+\infty}^{-\infty} \vec{B} \cdot d\vec{x}}{\mu_0}$, where $\vec{B}$ is the net magnetic field due to both the rings at any point on the x-axis.

13. A point source 'S' is moving in air towards left with a constant velocity $\frac{v}{3}$, where v is the speed of sound in air. There are two detectors A and B. A is moving in air with a velocity $\frac{v}{20}$ towards left and B is moving in a medium with a velocity $\frac{v}{9}$ towards right as shown in the figure. The velocity of sound in the medium is $4v$ and frequency of sound emitted by the point source is f. Let λ_A is the wavelength detected by the detector A and λ_B the wavelength detected by the detector B. If $\lambda_B - \lambda_A = k\left(\frac{v}{3f}\right)$, then find the value of k.



14. Atoms of type A fly along the axis of a cylindrical pipe of radius R and collide with practically motionless atoms of type B. The kinetic energy of atoms A is equal to the threshold energy, so that a central impact forms a molecule AB, which then moves with velocity v. In the case of an off-centre impact, the reaction does not occur, i.e. the atoms collide elastically. In what minimum time (in sec) after the collision will atoms of type B be able to get from the axis of the cylinder to the wall of the cylinder? ($R = 10$ cm and $v = 40$ cm / s)

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SECTION-III
(MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-1 & List-II). The options for the correct match are provided as (A), (B),(C) and (D) out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

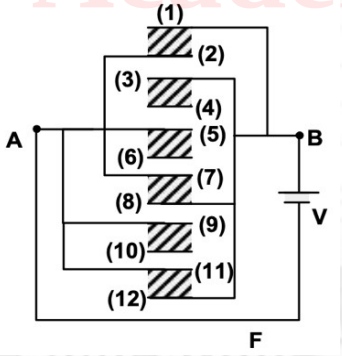
Section 3 (Max Marks: 12)

- This Section contains two (02) List-Match sets.
- Each List-Match set has TWO (02) Multiple-choice Questions
- Each List-Match set has two lists: List 1 and List 2
- List 1 has four entries (I), (II), (III), (IV) and List 2 has six entries (P), (Q), (R), (S), (T) and (U)
- Four options are given in each multiple choice question based on List 1 and List 2 and Only ONE of these four options satisfies the condition asked in the Multiple Choice Question.

Answer to each question will be evaluated according to the following marking scheme
Full Marks: +3 (If only the option corresponding to the correct combination is chosen))
Zero Marks: 0 (If none of the options is chosen (i.e. the question is unanswered)
Negative Marks: -1 (In all other cases)

Answer Q.15 and Q.16 by appropriately matching the list based on the information given in the paragraph

All identical capacitor plates each of area A are arranged such that adjacent plates are at d distance apart. Plates are connected to a source of emf V volts as shown in figure (dielectric constant in each shaded region is $k = 2$). Then



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	List - I		List - II
(I)	Charge on plate 3 is $\frac{10\varepsilon_0 AV}{Kd}$. Then the value of K is	(P)	zero
(II)	The heat rejected by the system is $\frac{26\varepsilon_0 AV^2}{Kd}$. Then the value of K is	(Q)	3
(III)	Equivalent capacitance between AB is $\frac{26\,A\varepsilon_0}{Kd}$. Then the value of K is	(R)	$\frac{13}{5}$
(IV)	Potential difference between the plates 4 and 7 is $\frac{KV}{5}$. Then the value of K is	(S)	$\frac{26}{5}$
		(T)	6
		(U)	$\frac{10}{3}$

15. Which of the following is correct?

A) I→U,II→Q,III→P,IV→Q **B)** I→T,II→P,III→Q,IV→U

C) I→U,II→T,III→Q,IV→P **D)** I→T,II→T,III→Q,IV→U

16. If the plates 4 and 5 are connected by a conducting wire. Then which of the following is correct.

A) I→U,II→S,III→R,IV→P **B)** I→T,II→S,III→R,IV→S

C) I→S,II→R,III→U,IV→P **D)** I→P,II→U,III→R,IV→P

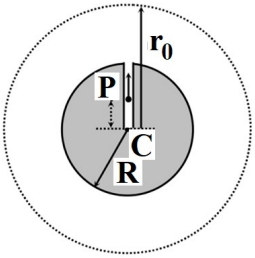
Answer Q.17 and Q.18 by appropriately matching the list based on the information given in the paragraph

A satellite is to be launched in a circular orbit of radius r_0 around a planet of radius R. For this two stage rocket is to be used. First stage rocket is used to lift the satellite vertically up to the desired height r_0 (measured from the center C of the planet), after launching it from a launch point P. Distance of launch point P from centre of the planet is r. Launch point P lies inside the tunnel as shown in the figure. Second stage rocket is used to impart it a tangential velocity so as to put it in a circular orbit. Let E_1 and E_2 are the energies delivered by the first and second stage of the rocket. Assume that the mass of the rocket is negligible with respect to the mass of the planet and atmospheric resistance to be zero. List-I gives 4 values of r_0 and r_L and List-II gives magnitude of some quantity.

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	List - I		List - II
(I)	$r_0 = R$ $r_L = \frac{R}{\sqrt{2}}$	(P)	$\frac{1}{2}$
(II)	$r_0 = \frac{3R}{2}$ $r_L = R$	(Q)	1
(III)	$r_0 = 4R$ $r_L = R$	(R)	$\frac{11}{6}$
(IV)	$r_0 = 4R$ $r_L = \frac{R}{\sqrt{2}}$	(S)	4
		(T)	6
		(U)	8

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17. The correct match for the time period of the satellite for the values of r_0 given in List-I in terms of T_0 is

(Take $T_0 = \frac{2\pi}{\sqrt{GM}} R^{3/2}$ and $\sqrt{6} = \frac{22}{9}$ if required)

A) I→Q,II→R,III→P,IV→U **B)** I→Q,II→P,III→U,IV→U

C) I→Q,II→T,III→U,IV→P **D)** I→Q,II→R,III→U,IV→U

18. The correct match for $\frac{E_1}{E_2}$ the values of r_0 and r_L given in List-I is

A) I→S,II→T,III→U,IV→P **B)** I→P,II→Q,III→T,IV→U

C) I→S,II→T,III→R,IV→P **D)** I→P, II→S, III→Q,IV→T

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CHEMISTRY **Max Marks: 62**

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 1 (Max Marks: 32)

- Section 1 contains eight questions.
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate the Answer to each question:
  Full Marks: +4 (If the answer is correct)
  Partial Marks: +3 (If all the four options are correct but only 3 are chosen)
  Partial Marks: +2 (If THREE or More options are correct but only 2 correct answers are chosen)
  Partial Marks: +1 (If TWO or More options are correct but only 1 correct answer is chosen)
  Zero Marks: 0 (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

19. W

The above image may stand for

A) chemical symbol for an *f*-block element

B) bond line notation for the most volatile isomers of pentane

C) one letter code for an essential α- amino acid called tryptophan

D) symbol for SI unit of power

20. In each of the following triads (set of 3 molecules) identify the one(s) where the middle member of the triad should be chosen as the correct response based on what is enquired against each triad.

A)	$CH(CN)_3$	$CH(NO_2)_3$	$CHCl_3$	strongest acid
B)	CH_3CONH_2	CH_3COOH	$CH_3COOC_2H_5$	highest melting point
C)	CCl_4	$SiCl_4$	$GeCl_4$	lowest boiling point
D)	H_2O	F_2O	Cl_2O	smallest bond angle

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21. Since NaN_3 is just as toxic as $NaCN$, it is no longer used in modern car airbags as a primary nitrogen gas generator. Instead solid guanidinium nitrate is used nowadays whose 1 mole decomposes completely to give 1 mole of solid soot (carbon), 2 moles of N_2 gas and 3 moles of water vapour. Identify the correct statement(s)

A) both NaN_3 and $NaCN$ undergo hydrolysis to produce weak acids that contain sp hybridised atom(s)

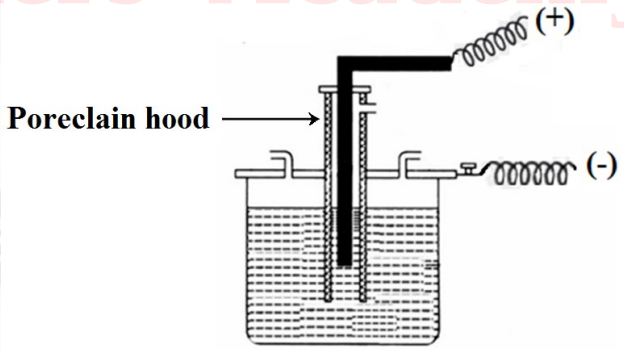
B) NaN_3 is used as a precursor in Curtius rearrangement

C) $NaCN$ is used as a catalyst in Benzoin condensation

D) a solution containing 1220 mg of guanidinium nitrate dissolved in 100 g of water starts to freeze at $-3.72^{\circ}C$ (minus three point seven two degree Celcius)

K_f for $H_2O = 1.86^{\circ}C$ kg mole^{-1}

22. Magnesium (atomic weight = 24) is extracted by the electrolysis of anhydrous and molten $MgCl_2$ (obtained from carnallite) using iron cell (container) as the cathode and graphite rod as the anode. The cross section of this electrolytic cell at the start of electrolysis is shown below.



Identify the correct statement(s)

A) to extract 3 kg of magnesium in an hour, more than 7 kiloamperes of constant and direct current is required.

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JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1
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B) anhydrous NaCl is added to the electrolyte to reduce melting temperature of the electrolyte and to improve electrical conductivity of the melt

C) porcelain hood provides a safe exit to chlorine gas thus preventing molten magnesium from recombining with chlorine

D) molten magnesium being denser than the electrolyte sinks to the bottom of the iron cell and is periodically tapped out

23. When subjected to Ruff degradation followed by treatment with conc. HNO_3 which of the following aldohexoses will lose their optical activity

A) D-Allose **B)** D-Altrose **C)** D-Glucose **D)** D-Gulose

24. Which aspect(s) of comparison is/are correctly listed (fully, not partially) in the chart drawn below

		N_2O_5	V_2O_5
A)	Physical appearance at room temperature	white solid	yellow-orange solid
B)	Nature	acidic	amphoteric with strong acidic bias
C)	Reaction with water	violent hydrolysis, even with cold water	slow hydrolysis, only with hot water
D)	Reaction with Ca, Δ	nitrogen gas is evolved	vanadium metal is obtained

25. Among the following step-growth polymers, select the one(s) which is/are polyamide(s)

A) Dacron **B)** Nylon-6, 6 **C)** Kevlar **D)** Glyptal

26. Which of the following statements about **aniline** is/are correct

A) it can help in the formation of a stable froth at the air-water interface during the concentration of sulphide ores by froth floatation method

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B) it reacts with benzene diazonium chloride in acidic medium to furnish a yellow coloured dye

C) it is obtained as a by product during glucosazone formation

D) it is used as a hardener for making ultra-filter paper

SECTION – II
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 2 (Max Marks: 18)

- Section two contains six (6) questions. The answer to each question is a NUMERICAL VALUE
- For each question enter the correct numerical value of the answer using the mouse and the on-screen virtual numerical keypad in the place designated to enter the answer. If the numerical values has more than two decimal places, truncate/round off the value to TWO decimal places.

Answer to each question will be evaluated according to the following marking scheme

Full Marks: +3 (If only the correct numerical value is entered)

Zero Marks: 0 (In all other cases)

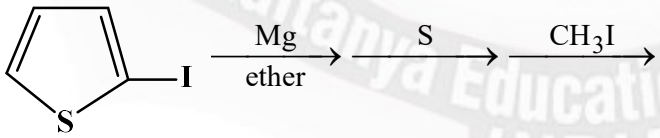
27. Victor Meyer test does not involve any carbocation rearrangement step, instead it works through a sequence of substitution and oxidation reactions. What is the maximum number of saturated alcohols containing exactly 6 carbons (M.F. $C_6H_{14}O$) which eventually show blue colour in Victor Meyer test proceedings (count stereoisomers separately)

28. What threshold pH must be maintained in a solution saturated with H_2S (0.1 M) and $10^{-3}M$ in Zn^{+2} to just prevent ZnS from precipitating.

$K_{sp}(ZnS) = 10^{-21}$

$Ka_1 \times K_{a_2}(H_2S) = 10^{-21}$

29. Calculate the molar mass of the end product (organic compound) in grams assuming all the steps occur with 100% yield. Atomic weights: H(1), C(12), Mg(24), S(32), I(127)



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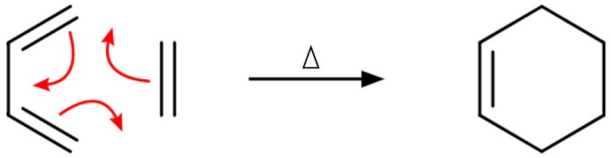
30. For the reaction $Br_2(\ell) \rightarrow Br_2(g)$

$\Delta H^\circ = 30$ kJ/mole

$\Delta S^\circ = 90$ J/mole. K

Assuming that ΔH and ΔS does not vary with temperature, calculate the boiling point of bromine on Kelvin scale.

31. In 1928, German chemist Otto Diels and his student Kurt Alder jointly proposed an extremely useful synthetic pathway for the construction of the cyclohexane framework. The simplest example of this reaction, which occurs in a single step, in shown below.



diene dieneophile

Study the reaction sequence shown below.

benzil + dibenzyl ketone + base $\xrightarrow{\Delta}$ **A**$(C_{29}H_{20}O)$
tetracyclone

A + maleic anhydride $\longrightarrow$ **B**

B + heat $\longrightarrow CO + H_2 +$ **C**

A + phenyl acetylene $\longrightarrow$ **D**

D + heat $\longrightarrow$ **E**

Calculate the degree of unsaturation in the hydrocarbon **(E)** produced at the end of the above reaction sequence.

32. In an ABCABC… packing of Giant Buckies (C-240), bucky babies – Chotu (C-32) and Motu (C-44) are doped in all tetrahedral and all octahedral voids respectively. In the unit cell thus created all the buckies along any one body diagonal are completely removed and Bucky Onions (C-240 with C-60 suspended inside it, with their centres coinciding) are fitted at either extremes of the body diagonal without distorting the size of the original unit cell. If the carbon atoms are assumed to be symmetrically distributed on the surface of all bucky balls, the number of carbon atoms finally present in the unit cell are _____.

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SECTION-III
(MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-1 & List-II). The options for the correct match are provided as (A), (B),(C) and (D) out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Section 3 (Max Marks: 12)

- This Section contains two (02) List-Match sets.
- Each List-Match set has TWO (02) Multiple-choice Questions
- Each List-Match set has two lists: List 1 and List 2
- List 1 has four entries (I), (II), (III), (IV) and List 2 has six entries (P), (Q), I, (S), (T) and (U)
- Four options are given in each multiple choice question based on List 1 and List 2 and Only ONE of these four options satisfies the condition asked in the Multiple Choice Question.

Answer to each question will be evaluated according to the following marking scheme

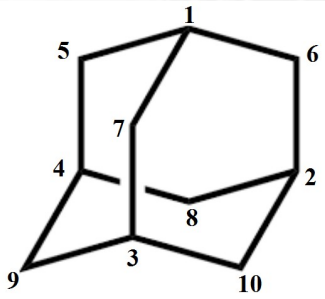
Full Marks: +3 (If only the option corresponding to the correct combination is chosen))

Zero Marks: 0 (If none of the options is chosen (i.e. the question is unanswered)

Negative Marks: -1 (In all other cases)

Answer Q.33 and Q.34 by appropriately matching the list based on the information given in the paragraph

Consider the structure shown below with each of its vertex marked by a natural number



Match **List-I** mentioning molecules derived from the above skeleton with **List-II** mentioning the property of these molecules

	List-I		List-II
I	assume one **P** atom each at vertex 1 to 4 and one **O** atom each at vertex 5 to 10	P	does not dissolve in pure water
II	assume one **As** atom each at vertex 1 to 4 and one **O** atom each at vertex 5 to 10	Q	dissolves in pure water but does not hydrolyse
III	assume one **N** atom each at vertex 1 to 4 and one methylene each at vertex 5 to 10	R	hydrolyses in excess aqueous alkali to form a monoanion
IV	assume the structure to be a hydrocarbon $(C_{10}H_{16})$ and then substitute any three of its tertiary hydrogens with three different halogen atoms	S.	hydrolyses in excess aqueous alkali to form a dianion
		T	hydrolyses in excess aqueous alkali to form a trianion
		U	exists as 16 different stereoisomers

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33. Which of the following options has the correct combination considering List-I and List-II

A) I, P **B)** I, Q **C)** II, S **D)** II, T

34. Which of the following options has the correct combination considering List-I and List-II

A) III, P **B)** III, R **C)** IV, P **D)** IV, U

Answer Q.35 and Q.36 by appropriately matching the list based on the information given in the paragraph

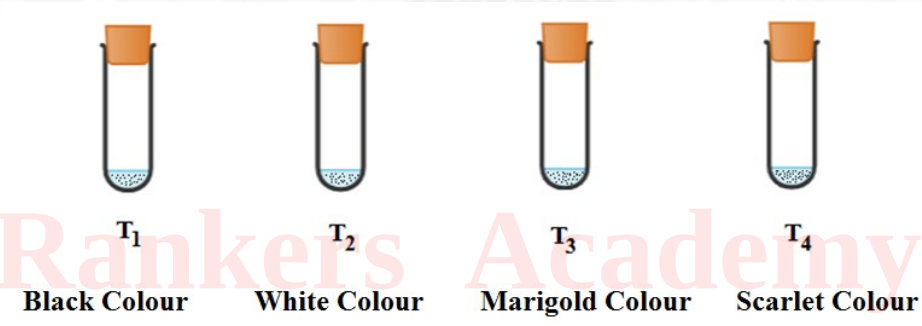
In 1899, a parcel saying 'Handle with care' was received at the following address:
Sherlock Holmes Detective Laboratory,
221B, Baker Street,

London,

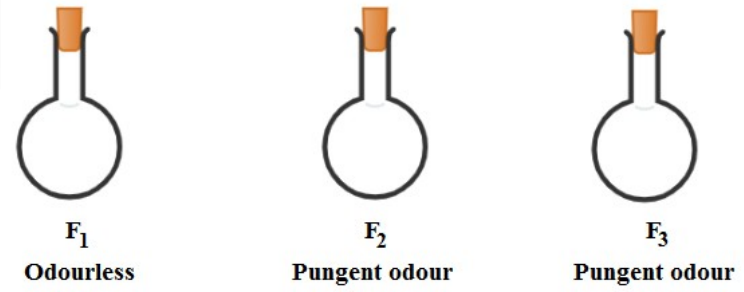
England.

The parcel was opened by Dr. John H. Watson, assistant to Sherlock Holmes. Inside, the following were found:

- **Test tubes containing solids with given colours (4 in number)**

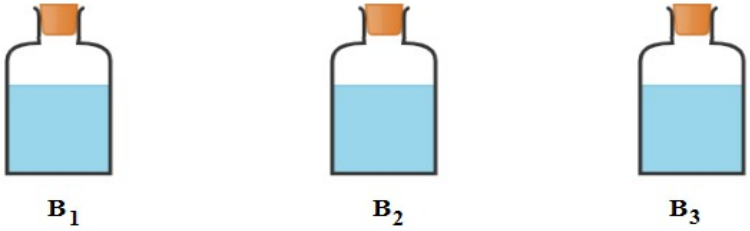


- **Flasks containing colourless gases: (3 in number)**



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JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1
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➢ **Bottles containing colourless liquids (3 in number)**



B_1 B_2 B_3

Labels: (only 4 labels were found, that too detached from their containers and lying on the floor of parcel. Rest 6 labels were missing)

$\boxed{K_2Cr_2O_7(s)}$ $\boxed{HgI_2(s)}$ $\boxed{H_2S(g)}$ $\boxed{NH_4OH(l)}$

Watson set about experimenting with the contents of the parcel in the laboratory. Here are the excerpts:

1. T_1 and T_4 were found not to dissolve in water. T_2 and T_3 were much soluble in water. Also, T_2 and T_3 imparted lilac colour to the Bunsen flame.
2. T_4 dissolves in an aqueous solution of T_2 due to complexation. The complex formed reacts with B_3 to form a dark-brown precipitate.
3. Both B_1 and T_3 on strong heating evolve gas F_1.
4. $T_1 + B_1 \xrightarrow{\Delta} PbSO_4 \downarrow$

   $T_1 + B_2 \xrightarrow{\Delta} PbSO_4 \downarrow + F_2 + H_2O$

   $B_1 + T_2(aq) \rightarrow I_2 \downarrow$

   $B_1 + F_3 \rightarrow S \downarrow$ (colloidal)

   $T_3 + B_1 + B_2 \rightarrow CrO_5$ (blue, stable in ether)

   $T_3(aq) + F_2 \rightarrow Cr_2(SO_4)_3$ (green)
5. Mixing equal moles of B_2 and B_3 results in neutralization due to which a salt is formed having $pH < 7$.

While Watson was trying to make some sense of all these observations, Sherlock Holmes entered the lab. He smiled at Watson who looked bewildered with 'papers' in his hands. Sherlock Holmes took the piece of paper wherein the observations were scribbled, took a deep look at them for a minute, and there it was- Holmes spelled out each of the ten unknowns in quick succession while Watson was dazed at his spontaneity. 'How did you manage to think so fast' asked Watson; to which Sherlock Holmes replied (taking a deep puff off his cigar and blowing out the smoke)- "Elementary, dear Watson".

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	List-I		List-II
I	B_1	P.	KI
II	F_2	Q.	$K_2Cr_2O_7$
III	B_3	R.	SO_2
IV	T_2	S.	H_2S
		T.	H_2O_2
		U.	NH_4OH

35. Which of the following options has the correct combination considering List-I and List-II

A) I, U **B)** I, P **C)** II, R **D)** II, S

36. Which of the following options has the correct combination considering List-I and List-II

A) III, U **B)** III, T **C)** IV, Q **D)** IV, S

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MATHEMATICS

Max Marks: 62

SECTION – I
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 1 (Max Marks: 32)

- Section 1 contains eight questions.
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate the Answer to each question:
  Full Marks: +4 (If the answer is correct)
  Partial Marks: +3 (If all the four options are correct but only 3 are chosen)
  Partial Marks: +2 (If THREE or More options are correct but only 2 correct answers are chosen)
  Partial Marks: +1 (If TWO or More options are correct but only 1 correct answer is chosen)
  Zero Marks: 0 (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

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37. Consider ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and parabola $y^2=2x$. They intersect at points P and Q in the first and fourth quadrants respectively. Tangents to the ellipse at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S. Latus rectum of parabola is L_1L_2 and latus rectum of ellipse is L_3L_4 where L_1 and L_3 lie in the first quadrant. Lines L_1L_4 and L_2L_3 intersect at point M(k,0). Which of the following options is/are true?

A) Area of quadrilateral PRQS is $15\sqrt{3}$ sq. units

B) $k=\frac{3\sqrt{5}+2}{7}$

C) Area of quadrilateral $L_1L_2L_4L_3$ is $\frac{7}{6}\left(2\sqrt{5}-1\right)$ sq. units

D) Equation of circle touching parabola at L_1 and passing through its vertex is $2x^2+2y^2-7x+y=0$

38. Let $f(x)=\sin(\cos^{-1}(\tan(\cos^{-1}x)))$. If α be a real solutions of the equation $f\left(f\left(x\right)\right)=f\left(x\right)$, then the value of $\left[\frac{\alpha}{2}\right]$ can be ([] denotes greatest integer function)

A) 1 **B)** 0 **C)** -1 **D)** 2

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39. Let $U=\{10,11,12,13,14,15,16,17,18,19,20\}$. Three sides of a triangle are selected from U, then which of the following statement(s) is/are incorrect

A) Number of triangles that can be made = 364

B) Number of triangles that can be made = 285

C) Number of isosceles triangles but not equilateral triangle that can be made = 130

D) Number of isosceles triangles that can be made = 260

40. Starting at $(-3,-3)$ a bug moves in the co-ordinate plane via a sequence of steps, each of length one, each step is left, right, up or down all four being equally likely. If P_n is the probability that object reaches $(-1,-1)$ first time in 'n' steps, then which of the following is/are CORRECT?

A) $P_4=\frac{3}{128}$ **B)** $P_5+P_6=\frac{3}{128}$ **C)** $P_4+P_6=\frac{3}{64}$ **D)** $P_4+P_5=\frac{3}{64}$

41. Vedant says to Chidvilas, "I am thinking of a polynomial whose roots are all positive integers. The polynomial has the form $P(x)=2x^3-2ax^2+(a^2-81)x-c$ for some positive integers a and c. Can you tell me the values of a and c?''

After some calculations, Chidvilas says, "There is more than one such polynomial."

Vedant says, "You're right. Here is the value of a." He writes down a positive integer and asks, "Can you tell me the value of c?"

Chidvilas says, "There are still two possible values of c."

What are the possible values of c?

A) 216 **B)** 220 **C)** 224 **D)** None of the above

42. Consider the sets

$A=\{(x,y)\ |x^2+4(y-a)^2-4=0,x,y,a\in R\}$

$B=\{(x,y)\ |x^2-2y=0,x,y\in R\}$

Which of the following is/are true?

A) for all $a\in\left[3,\frac{21}{5}\right], A\cap B\neq\phi$ **B)** If $A\cap B\neq\phi$ then $a\in\left[-1,\frac{17}{8}\right]$

C) If $a\in(0,2)$, then $A\cap B=\phi$ **D)** For all $a\in\left[3,\frac{21}{5}\right], A\cap B=\phi$

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IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023

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43. If three planes $P_1 = 2x + y + z - 1 = 0, P_2 = x - y + z - 2 = 0$ and $P_3 = \alpha x - y + 3z - 5 = 0$ intersects each other at point P on XOY plane and at point Q on YOZ plane, where O is origin then identify the correct statement(s)?

A) The value of α is 4

B) Straight line perpendicular to plane P_3 and passing through P is $\frac{x-1}{4} = \frac{y+1}{-1} = \frac{z}{3}$.

C) The length of projection of $\overrightarrow{PQ}$ on x-axis is 1

D) Centroid of the triangle OPQ is $\left(\frac{1}{3}, \frac{-1}{2}, \frac{1}{2}\right)$

44. If f is an odd continuous function in $[-1,1]$ and differentiable in $(-1,1)$ then which of the following statement(s) is (are) correct?

A) $f'(a) = f(1)$ for some $a \in (-1,0)$

B) $f'(b) = f(1)$ for some $b \in (0,1)$

C) $n(f(\alpha))^{n-1} f'(\alpha) = (f(1))^n$ for some $\alpha \in (-1,0)$ and $\forall\ n \in N$

D) $n(f(\beta))^{n-1} f'(\beta) = (f(1))^n$ for some $\beta \in (0,1)$ and $n \in N$

SECTION – II
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**

Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 2 (Max Marks: 18)

- Section two contains six (6) questions. The answer to each question is a NUMERICAL VALUE
- For each question enter the correct numerical value of the answer using the mouse and the on-screen virtual numerical keypad in the place designated to enter the answer. If the numerical values has more than two decimal places, truncate/round off the value to TWO decimal places.

Answer to each question will be evaluated according to the following marking scheme

Full Marks: +3 (If only the correct numerical value is entered)

Zero Marks: 0 (In all other cases)

45. Let $I_r = \int_a^{\pi+a} \left| \frac{1}{r} \sin x + \frac{1}{r+1} \cos x \right| dx$, where $a \in R$ and $r \in N$, then $\lim_{n\to\infty} \sum_{r=1}^{n} \left(I_r^2 - \frac{8}{(r+1)^2} \right)$ is equal to_____.

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IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 SINGARAJU VENKAT KOUNDINYA RANK 1
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46. Let $S_n = \sum_{k\geq 0} \binom{n+k}{2k} 2^{n-k}$ for $n \geq 3,\ n \in N$.

Also, let $2S_n = a\,S_{n-1} + b\,S_{n-2}$ for $n \geq 5,\ (a, b \in R)$

Then $a^2 + b^2 =$

47. Let $P(r)$ denote the number of points (x, y) (where $x, y \in I$) which lie inside or on the boundary of the region enclosed by the curve whose joint equation is

$|xy| - r(|x| + |y|) + r^2 = 0, r \in N$ and if $\lim_{n\to\infty} \frac{(\sum_{r=1}^{n} P(r)) - \lambda n^3}{an^2 + bn + c} = \frac{1}{2}$ then find the value of $a\lambda$

48. Let a_n be an arithmetic sequence such that arithmetic mean of $a_1, a_3, a_5, \ldots\ldots, a_{97}, a_{99}$ is 1. Find the value of $\left| \sum_{r=1}^{50} (-1)^{\frac{r(r+1)}{2}} . a_{2r-1} \right|$

49. If $\alpha^2 - 5\alpha + 2 = 0$ and $\beta^2 - 5\beta + 2 = 0, \alpha \neq \beta$ then $\frac{4\alpha + \beta^5}{10\beta^2}$ is equal to _____.

50. Let circular arc $\overset{\frown}{AC}$ and $\overset{\frown}{BC}$ have centers at $B(-\alpha, 0)$ and $A(\alpha, 0)$ respectively and equation of circle which touches both arcs $\overset{\frown}{AC}$ and $\overset{\frown}{BC}$ and line AB is $(x-a)^2 + (y-b)^2 = r^2, r > 0$, if length of arc $\overset{\frown}{BC} = 8\pi$, then the value of $|a + b + r - \alpha|$ is

(Point C lies, above x-axis)

SECTION-III
(MATCHING LIST TYPE)

This section contains 4 questions, each having two matching lists (List-1 & List-II). The options for the correct match are provided as (A), (B),(C) and (D) out of which ONLY ONE is correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

Section 3 (Max Marks: 12)

- This Section contains two (02) List-Match sets.
- Each List-Match set has TWO (02) Multiple-choice Questions
- Each List-Match set has two lists: List 1 and List 2
- List 1 has four entries (I), (II), (III), (IV) and List 2 has six entries (P), (Q), I, (S), (T) and (U)
- Four options are given in each multiple choice question based on List 1 and List 2 and Only ONE of these four options satisfies the condition asked in the Multiple Choice Question.

Answer to each question will be evaluated according to the following marking scheme

Full Marks: +3 (If only the option corresponding to the correct combination is chosen))

Zero Marks: 0 (If none of the options is chosen (i.e. the question is unanswered)

Negative Marks: -1 (In all other cases)

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Answer Q.51 and Q.52 by appropriately matching the list based on the information given in the paragraph

Let $\phi(x)=3f\left(\frac{x^2}{3}\right)+f(3-x^2)\ \forall\ x\in(-3,4)$ where $f''(x)>0\ \ \forall\ x\in(-3,4)$,

$$g(x)=(a^2-3a+2)\left(\cos^2\frac{x}{4}-\sin^2\frac{x}{4}\right)+(a-1)x+\sin 1$$

Match List-I with List-II and select the correct answer using the code given below the list.

	List - I		List - II
(P)	$\phi(x)$ is increasing in	(1)	$(0,1)\cup(1,4)$
(Q)	$\phi(x)$ is decreasing in	(2)	$\left(-\frac{3}{2},0\right)\cup\left(\frac{3}{2},4\right)$
(R)	The set of values of a for which $g(x)$ does not possess critical point is	(3)	$\left(-3,-\frac{3}{2}\right)\cup\left(0,\frac{3}{2}\right)$
(S)	The set of values of a for which $g(x)$ possess critical point is	(4)	$(-\infty,0]\cup\{1\}\cup[4,\infty)$

51. Which of the following is the only CORRECT combination

A) P, 3 **B)** R, 1 **C)** P, 1 **D)** S, 3

52. Which of the following is the only INCORRECT combination

A) P, 2 **B)** R, 1 **C)** Q, 3 **D)** Q, 4

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Answer Q.53 and Q.54 by appropriately matching the list based on the information given in the paragraph

Consider $P(x) = 2x^3 + x^2(3-7a) + x(3a^2 - 5a) + (3a^2 + 2a - 1)$.

Match List-I with List-II and select the correct answer using the codes given below the list.

	List - I		List - II
(P)	Number of positive value(s) of 'a' for which $P(x) = 0$ has exactly two equal roots, is	(1)	3
(Q)	Number of positive value(s) of 'a' for which all roots of $P(x) = 0$ are negative, is	(2)	1
(R)	Absolute value of sum of integral values of 'a' for which one root of $P(x) = 0$ is greater than -1 and other root is less than -1, is	(3)	2
(S)	Number of integral values of 'a' for which all roots of $P(x) = 0$ lie in $[-1, 2]$, is	(4)	0

53. Which of the following is the only CORRECT combination

A) P, 2 **B)** P, 3 **C)** Q, 1 **D)** Q, 2

54. Which of the following is the only INCORRECT combination

A) Q, 4 **B)** R, 1 **C)** S, 3 **D)** Q, 3

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